ACADEMIC CENTRAL

PRACTICE EXAMINATION

N6

MATHEMATICS N6

PRACTICE EXAM

MARKS: 100 **TIME: 3 hours**

NATED Engineering Studies, N6. Partial differentiation and small changes, integration techniques, areas, volumes and centroids, second moments of area, first-order and second-order differential equations. A formula sheet is attached.

Written by Academic Central for practice, following the NATED (Report 191) syllabus for N6. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL the intermediate steps.
5. Questions may be answered in any order, but subsections of questions must be kept together.
6. A formula sheet is attached at the end of this question paper.
7. Round off answers to THREE decimal places where necessary, unless stated otherwise.
8. Write neatly and legibly.

QUESTION 1: PARTIAL DIFFERENTIATION

1.1 Given $z = x^2y^3 - 3x \sin y$, determine:

1.1.1 $\dfrac{\partial z}{\partial x}$ (2)

1.1.2 $\dfrac{\partial z}{\partial y}$ (2)

1.1.3 $\dfrac{\partial^2 z}{\partial y\, \partial x}$ (2)

1.2 The volume of a cylinder is $V = \pi r^2 h$. The radius increases from 5 cm to 5,1 cm and the height decreases from 12 cm to 11,8 cm. Use partial differentiation to calculate the approximate change in volume. (6)

[12]

QUESTION 2: INTEGRATION

Determine:

2.1 $\int \left(3x^2 - \dfrac{4}{x} + e^{2x}\right) dx$ (3)

2.2 $\int \sin 3x \cos 2x \, dx$ (4)

2.3 $\int x\, e^{3x} \, dx$ (4)

2.4 $\int \dfrac{5x + 1}{(x - 1)(x + 2)} \, dx$ (6)

2.5 $\int \cos^3 x \, dx$ (4)

2.6 $\int \dfrac{1}{x^2 + 9} \, dx$ (3)

[24]

QUESTION 3: AREAS, VOLUMES AND CENTROIDS

3.1 Calculate the area enclosed between the curve $y = x^2$ and the line $y = 2x$. Make a neat sketch showing the area. (5)

3.2 The area bounded by the curve $y = x^2$, the x-axis and the lines $x = 0$ and $x = 2$ is rotated about the x-axis. Calculate the volume generated. (5)

3.3 Calculate the coordinates of the centroid of the area described in QUESTION 3.2. (8)

[18]

QUESTION 4: SECOND MOMENT OF AREA

4.1 Use integration to show that the second moment of area of a rectangle with breadth b and height h about its base is $I = \frac{bh^3}{3}$. Make a neat sketch showing a representative strip. (6)

4.2 Calculate the second moment of area about the base of a rectangle 6 cm wide and 4 cm high. (3)

4.3 Calculate the radius of gyration of the rectangle in QUESTION 4.2 about its base. (3)

[12]

QUESTION 5: FIRST-ORDER DIFFERENTIAL EQUATIONS

5.1 Solve the differential equation $\frac{dy}{dx} = 2xy$, given that $y = 3$ when $x = 0$. (6)

5.2 Determine the general solution of $\frac{dy}{dx} + 2y = e^{-x}$. (8)

[14]

QUESTION 6: SECOND-ORDER DIFFERENTIAL EQUATIONS

6.1 Determine the general solution of $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$. (4)

6.2 Determine the general solution of $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$. (3)

6.3 Determine the particular solution of $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 10$, given that $y = 3$ and $\frac{dy}{dx} = 0$ when $x = 0$. (13)

[20]

TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \; n \neq -1$

$\int \frac{1}{x} \, dx = \ln |x| + C$

$\int e^{ax} \, dx = \frac{1}{a} e^{ax} + C$

$\int \sin ax \, dx = -\frac{1}{a} \cos ax + C$

$\int \cos ax \, dx = \frac{1}{a} \sin ax + C$

$\int \frac{1}{x^2 + a^2} \, dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$

$\int u \, dv = uv - \int v \, du$

$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$

$\sin^2 x + \cos^2 x = 1$

$\delta z \approx \frac{\partial z}{\partial x} \delta x + \frac{\partial z}{\partial y} \delta y$

$A = \int_a^b (y_1 - y_2) \, dx$

$V_x = \pi \int_a^b y^2 \, dx$

$\bar{x} = \frac{\int xy \, dx}{\int y \, dx} \; ; \quad \bar{y} = \frac{\frac{1}{2} \int y^2 \, dx}{\int y \, dx}$

$I = \int r^2 \, dA \; ; \quad k = \sqrt{\left(\frac{I}{A}\right)}$

$\frac{dy}{dx} + Py = Q$: integrating factor $e^{\int P \, dx}$

Real, different roots: $y = Ae^{m_1 x} + Be^{m_2 x}$

Equal roots: $y = (A + Bx)e^{mx}$

Complex roots $m = \alpha \pm j\beta$: $y = e^{\alpha x}(A \cos \beta x + B \sin \beta x)$